

**MATERIAL SAFETY DATA SHEET  
CHLOROauric ACID SOLUTION**

**(HYDROGEN TETRACHLORO aurATE SOLUTION)**

**1. PRODUCT AND COMPANY IDENTIFICATION**

|              |                         |
|--------------|-------------------------|
| Trade Name   | <b>CHLOROauric ACID</b> |
| Product Type | Substance               |

**2. COMPOSITION/INFORMATION ON THE COMPONENTS**

|                     |                            |
|---------------------|----------------------------|
| Product Formal Name | Chloroauric acid solution  |
| CAS Number          | 16903-35-8                 |
| IUPAC name:         | hydron;tetrachlorogold(1-) |

**3. HAZARD IDENTIFICATION**

May cause severe burn. Harmful if swallowed. May result in burning sensation and coughing. May cause long term adverse effects.

**4. FIRST AID MEASURES**

|                        |                                                                                                                                                                                                                                                                 |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| First Aid              | Inhalation will result in burning sensation. May cause eye and skin burns. Flush eyes immediately with plenty of water for 10 minutes. Call for a physician.                                                                                                    |
| First Aid – Inhalation | Remove from exposure and allow to rest in fresh air. If breathing is stopped, give artificial respiration, if breathing difficult, give oxygen. Maintain an open airway. Loosen tight clothing such as collar, tie, belt, etc. Get immediate medical attention. |

|                       |                                                                                                                                                                                                                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| First Aid – Ingestion | Rinse mouth with plenty of water. Remove victim to fresh air. If material is swallowed give small quantity of water. Do not induce vomiting. Never give anything by mouth to an unconscious person. If vomiting occurs, head should be kept low to avoid vomiting enter the lungs. Seek medical attention. |
| First Aid – Eyes      | Flush eyes with water for 15-30 minutes lifting upper and lower eye lids. Remove contact lenses/glasses while flushing. Get medical attention immediately.                                                                                                                                                 |
| First Aid – Skin      | Remove contaminated clothing and wash skin with copious amount of soap and water. Wash clothing and shoes thoroughly before reuse. Get medical attention.                                                                                                                                                  |

## **5. FIRE FIGHTING MEASURES**

It is not a fire or explosion hazard. Wear positive pressure self-contained breathing apparatus and protective clothing. Wash contaminated clothing thoroughly with water before removing it, or wear gloves. Use extinguisher such as water, carbon dioxide, foam, dry chemical, etc. for the surrounding fire.

## **6. ACCIDENTAL RELEASE MEASURES**

Evacuate surrounding areas. Do not walk-through split material. Do not breathe vapor or mist. Wear full protective clothing and self-contained breathing apparatus while using. Ventilate area of leak. Avoid generating dust.

## **7. HANDLING AND STORAGE**

|          |                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Handling | Good industrial hygiene practices should be observed. Wash hands before eating, drinking, smoking or applying cosmetics and also at the end of the work shift. Use only under conditions of good local ventilation. Avoid contact with eyes, skin and clothing. Avoid generating or breathing dust. Avoid dust generation. Provide a safety shower and eye wash within the reach of the user. |
| Storage  | Keep in a cool, dark, dry place with adequate ventilation. Store in a tightly closed corrosive resistant container. Protect against physical damage.                                                                                                                                                                                                                                          |

## **8. EXPOSURE CONTROLS/PERSONAL PROTECTION**

Personal respiratory protection, gloves and industrial safety glasses should be worn. Wear clean/protective clothing like Lab Coat and Apron, boots/safety shoes, etc. to prevent exposure. Local exhaust ventilation is recommended in order to control dust, mist, vapor or fumes. Eye wash and safety drench shower should be within the reach of the user. Wash hands, forearms and face before handling any chemical products, before eating.

## **9 PHYSICAL AND CHEMICAL PROPERTIES**

|                      |                                                      |
|----------------------|------------------------------------------------------|
| Appearance           | Liquid Orange                                        |
| Odour                | Not available                                        |
| pH                   | <1                                                   |
| Boiling Range/Point  | Not available                                        |
| Melting Point        | Not available.                                       |
| Flash Point          | Not applicable                                       |
| Auto-flammability    | Not flammable                                        |
| Explosive Properties | Contact with Ammonia may produce explosive compounds |
| Oxidizing Properties | Not available                                        |

## **10. STABILITY AND REACTIVITY**

Possible ignition of flammable liquids and vapors. Under normal conditions of storage and use, hazardous decomposition productions should not be produced. Attacks many metals producing highly flammable hydrogen gas which can form explosive mixtures with air.

## **11. TOXICOLOGICAL INFORMATION**

No toxicological information available. However, it is harmful if swallowed. Eye contact, skin contact and inhalation may cause irritation.

## **12. ECOLOGICAL INFORMATION**

No ecological information available.

### 13. DISPOSAL

Return the accumulated waste material to the refinery for metal recovery, or dispose of in accordance with local and national regulations. Follow local disposal laws and regulations.

### 14. TRANSPORT INFORMATION

UN 3264, Class 8, Packing Group-II  
Corrosive material, acidic, inorganic, no's.

### 15. REGULATORY INFORMATION

Labelling information



Corrosive

R phrases

R23 Toxic by inhalation R22  
Harmful if swallowed.  
R20/22 – Harmful by inhalation and if swallowed  
R35 – Causes severe burns  
R53 – May cause long term adverse effects in the aquatic environment.

S phrases

S24/25 Avoid contact with skin and eyes.  
S26 In case of contact with eyes, rinse immediately with plenty of water and seek medical advice  
S28 After contact with skin, wash immediately with plenty of water  
S37/39 Wear suitable gloves and eye/face protection  
S45 In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible)

EINECS Number

2409484

EC Annex I Classification

C - Corrosive

### 16. OTHER INFORMATION

This information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process.

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## Reference Sources

Chemical Hazard Response Information System, Published by US Coast Guard.  
Institute of Occupational Safety and Health, USA.  
Registry of Toxic Effects of Chemical Substances, published by the National Oil and Hazardous Materials Technical Assistance Data System, Published by The Office of Water and Waste Management  
Environmental Protection Agency, USA  
NIOSH Occupational Safety and Health database, Published by the National Institute of Occupational Safety and Health, USA.  
RISKLIN, published by the National Chemical Inspectorate, Sweden  
Development and Reproductive Toxicology, The National Library of Medicine, USA  
HSELINE Health and Safety Executive Library and Information Service, published by the Health and Safety Executive, UK.  
TOXLINE, published by the Chemical Abstracts Service, USA  
CISDOC International Occupational Safety and Health Database, published by the International Labour Organization, Geneva.  
Toxicity, Bibliography, The National Library of Medicine, USA  
Environmental Mutagen Information, Oak Ridge National Laboratory, USA  
Environmental Teratology Information, Oak Ridge National Laboratory, USA  
BIOSIS, Published by Biological Abstracts Inc. USA  
Hazardous Materials Technical Center Abstracts, Department of the Army, Hazardous Materials Technical Center, USA.  
Toxicology Document and Data Depository, National Technical Information Service, USA.  
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International Pharmaceutical Abstracts, The American Society of Hospital Pharmacists, USA  
Handbook on Toxicology of Inorganic Compounds, Seiler, Sigel and Sigel  
Catalog of Teratogenic Agents, Shepard, 4<sup>th</sup> edition.  
Reproductive Hazards of Industrial Chemicals, Barlow and Sullivan  
Royal Society of Chemistry, Chemical Safety Data Sheets.  
Patty's Industrial Hygiene and Toxicology 3<sup>rd</sup> edition.  
Handbook on the Toxicology of Metals, Friberg, Nordberg and Vouk 2<sup>nd</sup> edition.  
Petherick's Handbook of Reactive Chemical Hazards, 5<sup>th</sup> edition  
Grant's Toxicology of the Eye, 3<sup>rd</sup> edition  
Merck Index 10<sup>th</sup> edition